DHAANUSH K

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PROFESSIONAL SUMMARY

AI & ML graduate (B.E., 2026) with hands-on experience building and evaluating deep learning classifiers and computer vision pipelines, including a peer-reviewed hybrid CNN + MLP model and a Vision Transformer-based image classifier. Comfortable across the ML lifecycle (data preprocessing, model training, evaluation) and full-stack engineering with authentication, automation and data pipelines. Strong grounding in Python, with SQL/MySQL, Java, Git and SDLC/Agile practices. Ranked 60th/150 teams at TechXcelerate, BITS Pilani.

TECHNICAL SKILLS

Languages: Python, SQL
EDA: NumPy, Pandas
ML & Deep Learning: Machine Learning, Deep Learning Classifiers, Neural Networks, TensorFlow, PyTorch, Scikit-learn
Computer Vision: OpenCV, Vision Transformers (ViT), Haar Cascade Classifiers, Data Augmentation, Transfer Learning
Web & Full-Stack: React, Node.js, Express.js, MongoDB, REST APIs, HTML/CSS
Applications & Libraries: Git/GitHub, Google Colab, Matplotlib/Seaborn, MySQL

EXPERIENCE

Image Processing Intern | *National Small Industries Corporation (NSIC), Chennai*

- Built a real-time Face Mask Detection System using Haar Cascade Classifiers and OpenCV, processing live video streams for face detection and mask classification.
- Developed image preprocessing pipelines (grayscale conversion, resizing, noise reduction, Canny/Sobel edge detection) to improve feature extraction for downstream object recognition tasks.

PROJECTS

Automated Accountability System with Real-Time PM Alerts | *MERN Stack*

- Built a full-stack MERN app automating team metrics tracking and project reporting; **Best Innovative Project Award, 2025**.
- Implemented JWT-based authentication with Role-Based Access Control (RBAC) and synced data in real time via the Google Sheets API.
- Automated deadline reminders and escalation emails using Nodemailer and Node-Cron, eliminating manual follow-up on missed submissions.

Wine Oracle: Quality Prediction with Neural Alchemy | *Deep Learning*

- Designed a hybrid MC-CNN + MLP model predicting wine quality across 14,853 samples (11 physicochemical features), achieving **87.6% (red) / 90.0% (white) accuracy**.
- Applied StandardScaler feature scaling, EDA, and Dropout (0.5) regularization to reduce overfitting and improve generalization.
- Co-authored resulting research as a peer-reviewed publication in the REST Journal on Data Analytics & AI (2024).

Medicinal Plant Classification using Vision Transformers | *Computer Vision*

- Built a ViT-based image classifier (PyTorch) over ~6,000 medicinal plant images, reaching **93.6% train / 93.2% test accuracy**.
- Engineered an end-to-end preprocessing pipeline (resizing, normalization, augmentation) and applied transfer learning with mixed-precision training for efficiency.

ACHIEVEMENTS & PUBLICATION

- **National Rank 60/150** — TechXcelerate 24-Hour Hackathon, BITS Pilani, Goa (AI/IoT track).
- Co-author, "*Wine Oracle: A Clandestine Journey into Quality Prediction with Neural Alchemy,*" REST Journal on Data Analytics and Artificial Intelligence, Vol. 3, Issue 2, 2024 (ISSN-indexed). DOI: 10.46632/jdaai/3/2/13.

EDUCATION

B.E, Computer Science and Engineering (AI & ML)

Vel Tech High Tech Dr. Rangarajan Dr. Sakunthala Engineering College, Chennai 2022 – 2026
CGPA: 7.9 / 10

Higher Secondary Certificate (HSC), State Board

The Ashram Matriculation Higher Secondary School 2022
Percentage: 73.16%

SSLC, State Board

The Ashram Matriculation Higher Secondary School 2020
Percentage: 91.4%